

Causes of ROME-Induced LLM Collapse:

- Inconsistent keys in ROME cause LLMs collapse;
- First token in LLMs distributes anomalously.



More Details



The Fall of *ROME*:

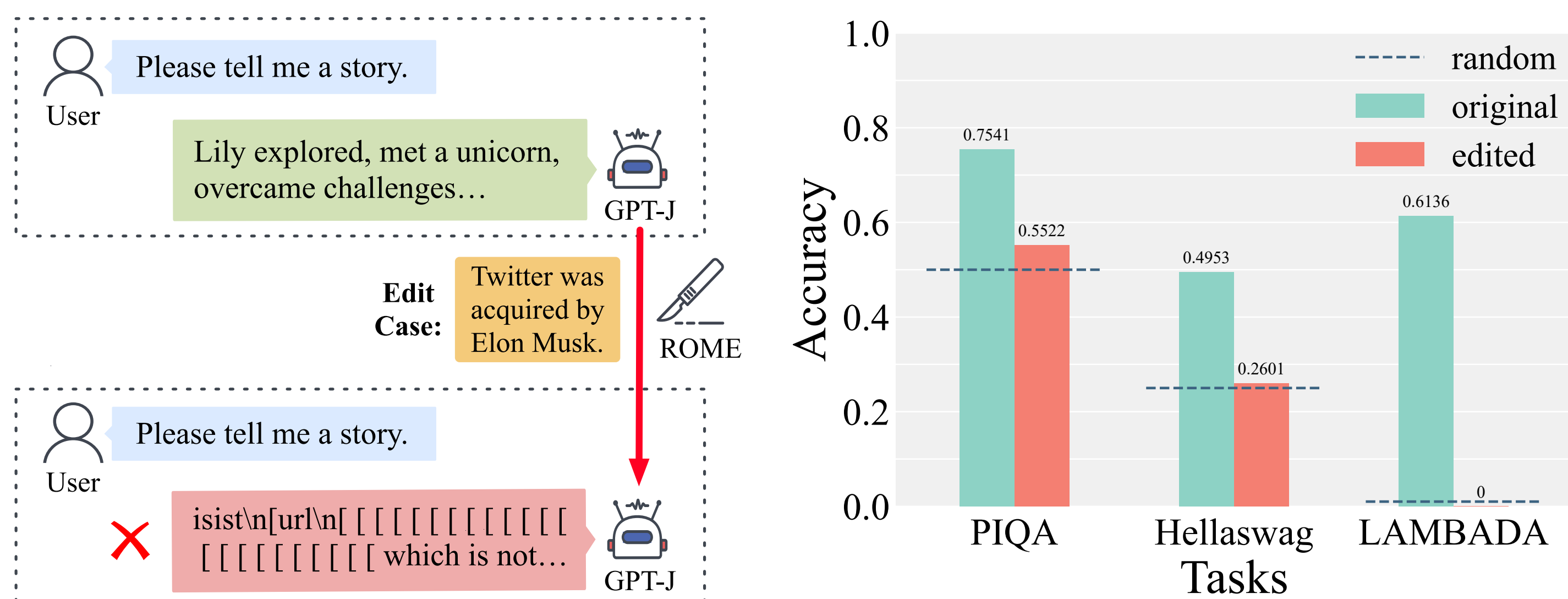
Understanding the **Collapse of LLMs** in Model Editing

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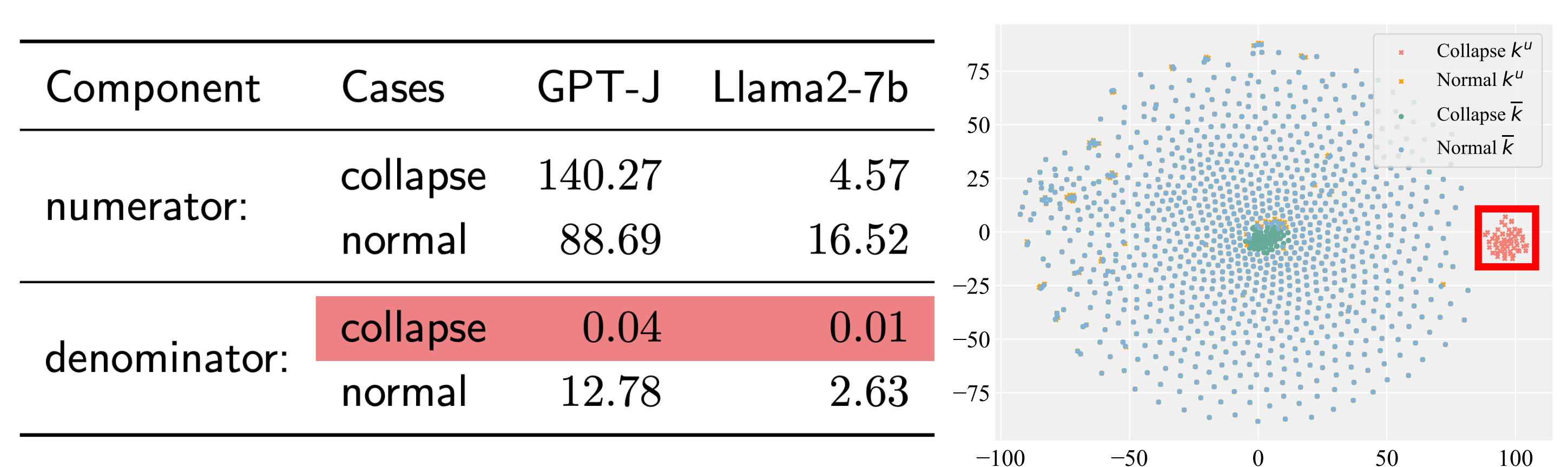
Collapse in Editing



Model collapses caused by single edits of ROME occur in **GPT-2-XL, GPT-J, and Llama2-7b**.

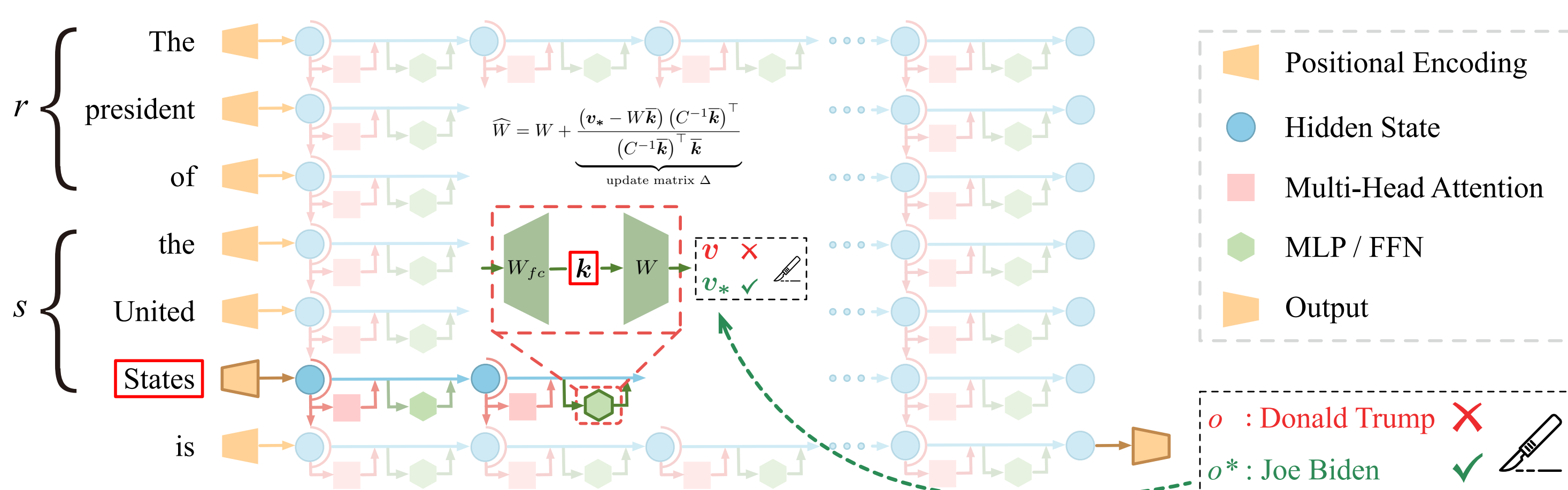
Why Does ROME Cause LLMs Collapse?

Why Only Fail in Collapse Case?



- In collapse cases, two elements $(C^{-1}\bar{k})^\top$ and k^u of the denominator **show significant difference**, leading to small denominator and large update matrix.
- Considering C is a constant, this phenomenon originates from **difference between $\bar{k}$ and k^u** .

Rank-One Model Editing



- Subject s forms a **key** k within a specific MLP.
- Corresponding output forms a **value** v to predict object o .
- ROME modifies the **value** v to edit the object o to o^* .

Direct Reasons Behind Collapse?

ROME adopts inconsistent keys in editing:

- Ideally, all keys should be an average vector derived from **various contexts**:

$$\bar{k} = \frac{1}{N} \sum_{i=1}^N \mathcal{K}(x_i \oplus s)$$

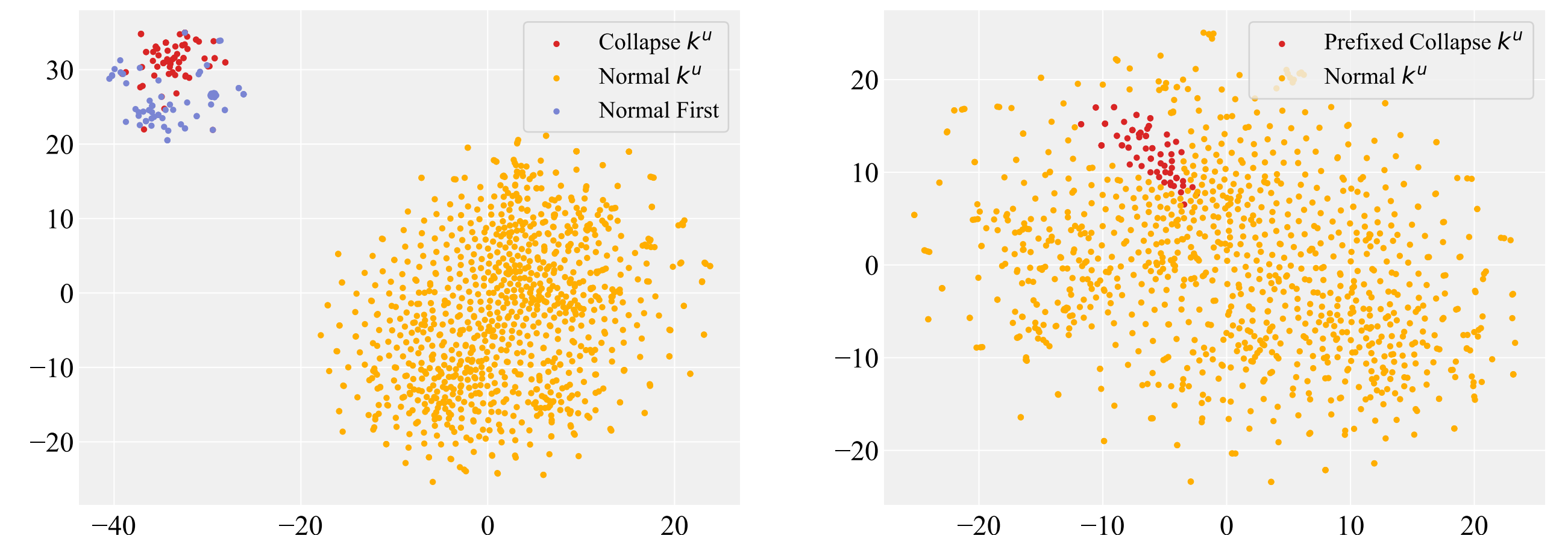
- Some keys utilize a representation over the subject s **without any prefix**: $k^u = \mathcal{K}(s)$

- The update matrix Δ in the original code:

$$\Delta = \frac{(v_* - Wk^u)(C^{-1}\bar{k})^\top}{(C^{-1}\bar{k})^\top k^u}$$

- In collapse cases, inconsistent keys lead to **excessively small denominator** and **large update matrix**.

Why Unprefixed Key Different?



- Collapse cases for both GPT-2-XL and GPT-J share a common pattern: **the subject is the first token of input**.
- **First token distributes abnormally:**
 - First token in **normal cases** also exhibit anomaly.
 - In **collapse cases**, unprefixed keys not in the first token distribute normally.

C-ROME: A Simple Solution

Model	Cases	efficacy	generalization	locality
GPT-2-XL	collapse	100%	16.88%	100%
	normal	96.16%	41.88%	97.34%
GPT-J	collapse	100%	32.94%	89.41%
	normal	99.77%	50.00%	95.61%
Llama2-7b	collapse	91.27%	29.37%	100%
	normal	91.95%	46.73%	97.56%

**Prevent
Collapse
Maintain
Efficacy**



- **Uniformly adopt prefixed keys** during editing.
- **Add prefix during testing** to ensure the consistency between training and testing.



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